Table S1. Parameter Explanations for the $PM_{2.5}$ Prediction Model. This table lists the input parameters used in the $PM_{2.5}$ concentration prediction model and provides a brief explanation of their impact on air quality.

Table S2. Air Quality Level Classification Based on the "Technical Regulations for Ambient Air Quality Index"

Fig.S1. Distribution of Stations in the Beijing-Tianjin-Hebei Region.

Table S1. Parameter Explanations for PM2.5 Prediction Model

Parameter	Explanation
AQI (Air Quality Index)	A single numerical indicator that integrates multiple pollutant concentrations (e.g., $PM_{2.5}$, PM_{10}, O_3) to represent overall air quality. It helps track general pollution levels and provides indirect insights into $PM_{2.5}$ trends.
CO (Carbon Monoxide)	Concentration of Carbon Monoxide, mainly from vehicle exhaust and combustion sources. CO often co-occurs with $PM_{2.5}$ in urban areas, indicating shared emission origins.
NO_2 (Nitrogen Dioxide)	A reactive nitrogen gas produced mainly by fossil fuel combustion. It is a precursor to secondary particulate formation and correlates with traffic-related $PM_{2.5}$ emissions.
O_3 (Ozone)	A secondary pollutant formed through photochemical reactions involving NO_x and VOCs. Ozone levels influence the oxidative environment and the formation of secondary $PM_{2.5}$.
PM_{10} (Particulate Matter 10)	Particulate matter with diameter ≤10μm, which includes $PM_{2.5}$. PM10 trends often move with PM2.5 and can indicate broader particle pollution.
$PM_{2.5}$（Particulate Matter 2.5）	Fine particulate matter with diameter ≤2.5μm. It poses significant health risks and is the core target variable in the prediction model.
SO_2 (Sulfur Dioxide)	A gaseous pollutant from industrial processes and coal combustion. SO_2 can react in the atmosphere to form sulfate particles, contributing to $PM_{2.5}$ levels.
t2m (2-meter Temperature)	Near-surface air temperature at 2 meters. It influences atmospheric mixing, stability, and chemical reaction rates, all of which affect $PM_{2.5}$ dynamics.
d2m (2-meter Dew Point Temperature)	Dew point temperature at 2 meters, reflecting ambient humidity. High dew points are associated with enhanced aerosol hygroscopic growth and secondary PM formation.
V10 (10-metre v-component of wind)	Meridional (north-south) wind component at 10 meters altitude. It affects pollutant transport and regional dispersion

	of $PM_{2.5}$.
u10 (10-metreu-component of wind)	Zonal (east-west) wind component at 10 meters altitude. Together with V10, it determines wind-driven horizontal movement of air pollutants.
tp (Total Precipitation)	Total precipitation rate in millimeters per hour. Rainfall helps remove $PM_{2.5}$ from the air through wet deposition.
sp (Surface Pressure)	Surface atmospheric pressure, indicating synoptic weather conditions. It influences boundary layer height and the vertical mixing of pollutants, thus affecting $PM_{2.5}$ accumulation.

Table S2. Air Quality Level Classification Based on the "Technical Regulations for Ambient Air Quality Index"

Level	PM2.5 concentration range (μg/m³)
Excellent	0-35
Good	36-75
Light Pollution	76-115
Moderate Pollution	116-150
Heavy Pollution	151-250
Severe Pollution	>250

Fig.S1. Distribution of Stations in the Beijing-Tianjin-Hebei Region